Simple Controls Remain Hard to Beat in Learned Multimodal Routing:
Five Controlled Studies and a Supported Residual Dynamics Result

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Abstract

Adaptive fusion is often motivated by the intuition that a learned router should down-weight unreliable inputs better than a fixed mixture. We tested this intuition in a sequence of prospectively specified studies built around Olympus, an auditable reference architecture for heterogeneous prediction. Four synthetic studies progressively added explicit reliability supervision, independently trained experts, cross-fitted routing, broader corruption coverage, and development-only route selection. A fifth study used the licensed EditCLEVR paired-image benchmark. Across all five studies, the registered learned routing candidate did not beat its strongest matched simple control on the primary endpoint. The successive primary effects were -0.0469 selective-accuracy points versus uniform fusion, -0.0125 versus semantic routing, -0.0682 accuracy points versus semantic routing, -0.00068 versus semantic routing on novel shift, and -0.00005 versus a development-selected EditCLEVR control. In contrast, a separately registered residual temporal predictor reduced clean mean-squared error from 0.07133 for the prior predictor to 0.01457, beat persistence by 0.01540 with a paired descriptive 95% interval of $[-0.01583, -0.01492]$, and retained corrupted-input recovery. The evidence supports mixed-corruption training, sample-dependent semantic routing in one observed corruption regime, and residual prediction for the synthetic dynamics task. It does not support learned routing as a general improvement, natural-data robustness, deployment, safety efficacy, or a full cognitive architecture. The contribution is a preserved sequence of falsifiable iterations showing where increasingly elaborate routing repairs failed, where one simpler architectural prior succeeded, and how negative outcomes can be reported without retroactive endpoint selection.

1 Introduction

Multimodal prediction systems must operate when some inputs are missing, corrupted, or less useful for the current example. Deep fusion can obscure which input drove a decision, while mixture-of-experts systems add their own gate-collapse and optimization failure modes [2, 11, 10]. Missing-modality studies also show that robustness is sensitive to the dataset, missingness pattern, and fusion design [7, 8]. Recent work has therefore explored explicit routing, reliability estimation, reconstruction, and sequential evidence selection [4, 6, 9].

The motivating hypothesis is attractive: if a router can infer input reliability for each example, it should outperform a uniform or fixed mixture under shift. The hypothesis is also easy to make unfalsifiable. A router can be changed after inspecting test outcomes; an oracle can be mistaken for a deployable method; a weak baseline can make any adaptive mechanism look useful; or an overall score can hide a clean-versus-shift tradeoff. This paper instead reports a sequence in which every failed primary outcome remained terminal for its registered protocol and every repair became a separately versioned successor.

Our main findings are:

1. Five learned-routing candidates failed their primary comparison, including a test on the public, CC-BY-4.0 EditCLEVR benchmark [5].

2. The failures persisted after the experiments added direct reliability labels, independently supervised expert banks, cross-fitting, corruption-diverse development data, and development-only selection among candidate routers.

3. Mixed-corruption training and semantic routing nevertheless produced large matched improvements in one synthetic corruption regime. These are narrower findings than a general routing claim and did not transfer uniformly to held-out shifts.

4. A residual temporal predictor passed every registered primary criterion, including a paired comparison with persistence. This positive result concerns synthetic dynamics, not the routing hypothesis or the full Olympus system.

The study is deliberately not presented as state of the art. The models are small, four studies are synthetic, the external study uses one benchmark, and the final two remote evidence archives were verified in their executing notebook but were not recovered into the repository. Those constraints are part of the result, not footnotes to be removed.

2 Architecture and claim boundary

Olympus is a reference architecture in which m component families emit states h_i. Each state is projected and normalized,

$$z_i = \mathrm{LayerNorm}(W_i h_i + b_i), \tag{1}$$

$$g = \mathrm{softmax}(r([z_1; \ldots; z_m])/\tau), \tag{2}$$

$$z = \mathrm{LayerNorm}\left(\sum_i g_i z_i\right), \tag{3}$$

and a shared decision head acts on z. The system also exposes leave-one-component-out states

$$z_{-i} = \mathrm{LayerNorm}\left(\frac{\sum_j g_j z_j - g_i z_i}{\max(1 - g_i, \epsilon)}\right), \tag{4}$$

so that component-removal behavior remains observable. A counterfactual regularizer can penalize divergence between the full and removal predictions, with the full prediction stop-gradient treated as the target.

The architecture distinguishes learned intrinsic risk q from a host-provided risk floor r_h. Their effective value is

$$r_{\text{eff}} = 1 - (1 - r_h)(1 - q). \tag{5}$$

For $q, r_h \in [0, 1]$, $r_{\text{eff}} - r_h = q(1 - r_h) \geq 0$. Learned evidence therefore cannot reduce the host risk floor. Host-provided legal-transition masks are applied after the learned scores, and a fixed threshold can force STOP. These are algebraic and software invariants. They are not evidence that the learned risk is calibrated, that the host risk is correct, or that deployment is safe. Selective prediction and conformal risk control provide important neighboring frameworks, but the invariant alone supplies neither coverage nor distribution-free risk control [3, 1].

The repository names six component families, but the studies below isolate routing and temporal mechanisms at reference scale. They do not validate six-family joint capability, the separately specified joint-embedding upscaler, production behavior, or any cognitive claim.

3 Sequential research design

Each iteration was opened only after the predecessor result had been recorded. Protocols fixed the question, controls, source-disjoint splits, budgets, endpoint, effect threshold, and protected-outcome chronology before confirmation. A failed study could motivate a new hypothesis, but its endpoint, data, and disposition were not rewritten.

Study	Registered repair	Strongest primary control	Primary disposition
v2	Counterfactual fusion and disagreement-conditioned risk	Exact uniform projected fusion	not supported
v3	Supervised reliability heads and shrinkage toward uniform	Semantic routing	not supported
v4	Cross-fitted expert banks and utility routing	Semantic routing	not supported
v5	Broader shift coverage and development-only candidate selection	Core-training semantic routing	not supported
EditCLEVR	Four routed paired-image experts	Best validation-selected simple control	not supported
v5 temporal	Residual state prediction	Prior v4 predictor and persistence	supported

Table 1: The iteration sequence. A new row is a new frozen hypothesis, not a reinterpretation of an earlier protected result.

All paired intervals reported below are fixed-seed percentile bootstrap intervals over five training seeds or five independently generated dataset clusters, depending on the protocol. With only five units, they are descriptive rather than asymptotically precise. Secondary outcomes cannot rescue a failed primary endpoint. Oracle routes are diagnostics that use privileged ground truth and are never counted as deployable candidates.

3.1 v2: controlled reliability shift

The first confirmatory study generated 1,408 source-disjoint units: 512 training, 128 development, and three protected sets of 256 for clean, corrupted, and out-of-distribution conditions. Five matched variants were trained for five seeds, giving 25 fits and 75 protected evaluations. The primary claim required the full mechanism to exceed uniform fusion by at least 0.02 clean accuracy and 0.03 corrupted selective accuracy, with positive paired interval lower bounds.

3.2 v3: direct reliability supervision

The successor asked whether reliability identification could repair routing. It added one reliability head per component, supervised those heads using known interventions, combined semantic and reliability evidence in log space, and shrank the result toward uniform. Reliability targets were labels during training and were not supplied at inference. Five matched variants, five seeds, 25 fits, and 75 protected evaluations were frozen. The primary endpoint was corrupted selective accuracy at 50% coverage against semantic routing, with a required effect of at least 0.02 and a positive interval lower bound.

3.3 v4: expert-bank diagnostic

The third iteration removed label-recoverable metadata shortcuts and introduced five independently supervised expert heads. Two-fold cross-fitting ensured that routing features for an example came from a bank not trained on that example. Missing-only and mixed missing-plus-perturbed training were paired on identical latent examples and budgets. Utility, semantic, fixed, uniform, corruption-proxy, best-single, and privileged oracle routes shared the same banks. The design contained 40 cells, 80 expert-bank fits, and 120 protected regime evaluations. The utility router had to beat the development-selected control by at least 0.02 on corrupted accuracy with a positive cluster interval lower bound.

3.4 v5: broader shifts and residual dynamics

The fourth synthetic iteration used fresh units and reserved a genuinely novel synthetic shift family for protected evaluation. Core training used the v4 corruption set. Broad training added token reversal, image inversion, and positive state bias during development; novel protected shifts used token relocation, checkerboard image occlusion, and negative time-varying state drift. Five dataset seeds and two training seeds yielded 20 cells, 40 bank fits, and 80 protected evaluations.

Candidate routing was selected using development clean, corrupted, and known-shift data by maximizing worst-regime and then mean accuracy under a fixed tie order. The registered routing comparison was broad selected routing minus core semantic routing on novel shift, with a required effect of 0.02 and corrupted non-regression.

The same protocol separately compared a prior temporal predictor, persistence, a direct MLP, and a residual MLP. The two new MLPs had the same 88→64→5 architecture, initialization, 500-update budget, and inputs. The residual model alone added the observed state to the predicted change. It had to reduce clean MSE by at least 10% relative to both the prior predictor and persistence, have a residual-minus- persistence interval upper bound below zero, and remain within 0.02 MSE of the prior model under corruption.

3.5 EditCLEVR paired-image study

EditCLEVR provides before-and-after CLEVR-style scenes with known object-factor interventions and controlled distribution shifts [5]. We pinned dataset revision `734efa9b0164ba742cd2c39b9c2945c494497c23`, whose local receipt accounts for 20,000 unique pair identifiers and 120,000 referenced files with no missing paths. The dataset is CC-BY-4.0.

The bounded study predicts the edit factor rather than claiming the benchmark's full scene-graph intervention metric. Shared convolutional encoders feed before-image, after-image, absolute-change, and paired-image experts. Learned, uniform, pair-only, and delta-only predictions come from the same checkpoint. The strongest simple control is selected on validation accuracy. Five optimizer seeds ran 15 epochs each; four protected splits produced 20 protected reports. The primary effect averages hard and CoGenT-shifted accuracy. It required a learned-minus-control mean of at least 0.02 with a positive interval lower bound, plus in-distribution non-regression.

4 Results

4.1 Learned routing did not win a primary comparison

Table 2 reports every routing primary outcome. None reached its registered effect threshold, and four point estimates favored the matched control. The EditCLEVR effect was effectively a tie: four seeds tied exactly and one was slightly worse.

Study	Endpoint	Candidate minus control	Descriptive 95% interval	Required
v2	Corrupt selective accuracy	-0.046875	$[-0.0953, 0.0016]$	$+0.030$
v3	Corrupt selective accuracy	-0.012500	$[-0.0281, 0.0031]$	$+0.020$
v4	Corrupt accuracy	-0.068164	$[-0.0771, -0.0589]$	$+0.020$
v5	Novel-shift accuracy	-0.000684	$[-0.00244, 0.00107]$	$+0.020$
EditCLEVR	Shifted accuracy	-0.000050	$[-0.00015, 0]$	$+0.020$

Table 2: Registered routing primary outcomes. The control is exact uniform fusion in v2, semantic routing in v3 and v4, core semantic routing in v5, and the validation-selected simple control for EditCLEVR.

The v2 full mechanism tied uniform fusion on clean accuracy exactly, but its corrupted selective point estimate was worse. Its gates had normalized entropy 0.922 and an effective component count of 4.42, so hard gate collapse was not the main explanation. Intrinsic risk was weakly aligned with the constructed corruption target, yielding a high risk Brier score of 0.741.

In v3, reliability supervision produced positive mean separation between intact and damaged positions under corruption, but that signal did not improve the task endpoint. This distinguishes failure to identify corruption from failure to convert a coarse reliability signal into useful task routing. A source can be damaged yet remain discriminative, or remain clean while being redundant.

In v4, the utility router lost to semantic routing in all five dataset clusters. Privileged diagnostics showed that useful expert signal still existed: on corrupted inputs, a task oracle exceeded uniform fusion by 0.1895 and a corruption oracle exceeded it by 0.1482. Those oracles use unavailable ground truth, so they identify headroom without constituting a deployable result.

In v5, broader coverage plus development-only candidate selection did not improve novel shift performance over the core semantic control, and it materially regressed on the original corruption regime by -0.05869 with interval $[-0.06387, -0.05068]$. This rejects the idea that the registered selection rule converted broader observed shifts into robust router choice.

On EditCLEVR, the learned router tied its selected control in four of five seeds and was worse by 0.00025 in one. Its mean in-distribution difference was -0.000067 with interval $[-0.00020, 0]$, which passed the registered non-regression condition but supplied no evidence of improvement.

4.2 Narrow secondary improvements

The failed primary routing claim does not imply that every intervention was useless. In v4, mixed missing-plus-perturbed training improved semantic-router corrupted accuracy from 68.32% to 95.74%, a matched increase of 27.42 percentage points with interval $[25.30, 29.28]$. On the same mixed-training banks, semantic routing exceeded learned fixed routing by 10.21 points, interval $[9.62, 10.76]$, and its own mean training gates by 11.54 points, interval $[10.27, 13.18]$. These outcomes show that input-dependent routing can help within the observed corruption regime.

The gain did not generalize uniformly. Semantic routing scored 91.99% on the v4 OOD split, below uniform at 97.88%, learned fixed at 98.64%, and the utility router at 99.55%. Selecting a post hoc hybrid after observing these protected outcomes would create a new hypothesis rather than repair the v4 result.

4.3 Residual temporal prediction passed its primary endpoint

The v5 residual temporal model was the only registered candidate in this sequence to pass all of its primary criteria. Table 3 reports the comparison.

Quantity	Result	Criterion
Residual clean MSE	0.014566	lower is better
Prior v4 predictor clean MSE	0.071332	residual $\leq 90\%$
Persistence clean MSE	0.029964	residual $\leq 90\%$
Reduction versus prior v4 predictor	79.58%	at least 10%
Residual minus persistence	-0.015398	interval upper < 0
Paired descriptive 95% interval	$[-0.015830, -0.014920]$	passed
Residual corrupted MSE	0.087415	no more than prior $+0.02$
Prior v4 predictor corrupted MSE	0.138464	passed

Table 3: v5 residual temporal primary result. The persistence MSE is derived from the reported paired difference and residual MSE. The result is specific to the frozen synthetic dynamics generator.

The skip connection is consistent with the target being easier to model as a change from the observed state. However, comparison with the prior v4 predictor also changes the head architecture and training budget. Only the direct-versus-residual comparison isolates the skip connection within the new head family, and no natural trajectory benchmark was tested.

4.4 Invariant and denominator checks

The v2 and v3 studies each completed 25 fits and 75 protected evaluations. v4 completed 40 cells, 80 bank fits, and 120 protected evaluations. v5 completed 20 cells, 40 bank fits, and 80 protected evaluations. EditCLEVR completed five training seeds and 20 protected reports. No failed or unfavorable study was deleted from the sequence. The v2 and v3 protected receipts reported zero host-risk-monotonicity and transition-legality violations. These counts establish completeness for the stated study plans, not external replication.

5 Why increasingly elaborate routing repairs failed

The sequence narrows several plausible explanations.

Reliability is not utility. v3 learned a corruption-related signal without improving task routing. Reliability labels describe an intervention, while the desired gate depends on class-conditional information, redundancy, calibration, and interactions among experts.

Useful experts do not imply a learnable router. The v4 task oracle revealed substantial headroom. Its access to true-class likelihood is exactly what a deployed router lacks. Cross-fitting removed direct in-sample leakage but did not make the utility target identify the correct expert under shift.

Coverage can trade one shift for another. Mixed-corruption training strongly repaired the observed corruption regime, yet semantic routing regressed on v4 OOD inputs. Broader v5 development coverage still did not improve the reserved novel shifts and harmed the original corruption comparison.

Selection adds another generalization problem. Choosing among routers on development data does not guarantee that the selected rule will transfer to a new intervention family. v5 froze selection before protected access, making that failure visible instead of silently selecting the protected winner.

Simple inductive bias can outperform learned arbitration. The residual temporal model embeds a direct assumption: predict change around the observed state. On the frozen dynamics task, this bias was more effective than adding another learned selection mechanism. This does not make residual prediction universally superior; it shows that a matched simple baseline should be treated as a first-class candidate.

6 Evidence integrity and reproducibility

The repository at `https://github.com/build-the-future-11/olympus-cognitive-architecture` contains the implementation, protocols, frozen configuration identities, historical states, tests, and manuscript source. Earlier v2, v3, and v4 evidence was recovered locally with source and prediction hashes. The v5 and EditCLEVR executing notebooks produced complete verifier reports: 80 and 20 prediction hashes respectively, recomputed aggregate means and paired deltas, checkpoint/source chronology checks, and zero cross-fit overlap where applicable. These verifiers establish provenance and arithmetic consistency, not an independent retraining.

The final v5 evidence archive had remote SHA-256:

`7dd4b614d2895331e2000a68d2ae988cf8badd54d2776be26857471a5ac391d6`

The EditCLEVR archive had SHA-256:

`a947b3f86afe71ff712e7f203fdfc421cb7c66d3a45e5e61d2d95f6ffeed13c1`

Neither archive was recovered from the ephemeral notebook working directory. The local repository therefore preserves their protocols, exact source, canonical result states, remote archive digests, and verifier summaries, but not the final raw remote predictions or checkpoints. This limitation prevents a claim that the public artifact alone can recompute every v5 and EditCLEVR number.

The publication package is generated without proprietary smoke records, checkpoints, or dataset payloads. EditCLEVR is referenced by pinned revision rather than redistributed. Software tests, type checks, package builds, and PDF compilation establish engineering integrity only. They do not add scientific samples or strengthen an effect estimate.

7 Limitations

Four routing studies use synthetic generators and small reference models. Synthetic interventions provide exact labels and controlled pairing but cannot represent the full dependence structure of natural multimodal data. EditCLEVR is public and licensed, but our study predicts edit factors and does not execute the benchmark's complete object-centric faithfulness protocol. Five seeds or five dataset clusters offer limited precision, and the intervals are descriptive. The same research program designed and verified the studies; there is no independent operator reproduction. The final remote archives are not locally available. No target-scale model, human evaluation, energy study, calibrated deployment, rights-reviewed six-family corpus, or safety intervention is evaluated. The clean synthetic classification tasks often saturate, which limits their ability to distinguish methods.

The result also does not establish that all learned routing must fail. It rejects five specific, frozen mechanisms under their stated data and budgets. Other targets, causal signals, architectures, or substantially larger datasets may behave differently. Testing them would require fresh hypotheses and protected examples rather than reuse of these outcomes for tuning.

8 Conclusion

Across five controlled studies, adding reliability supervision, expert-bank separation, cross-fitting, corruption coverage, and development-only route selection did not make the registered learned router outperform its strongest

simple matched control. The sequence does support two narrower conclusions: corruption-diverse training can matter greatly for the corruption families it covers, and input-dependent semantic routing can outperform a fixed mixture within one such regime. Neither finding generalized into a successful reserved-shift routing primary.

The residual temporal predictor provides a contrasting positive result. It passed a predeclared comparison with both the prior predictor and persistence while preserving corrupted-input performance. The practical lesson is not that complexity never helps. It is that adaptive routing, reliability estimation, and model selection each create a new generalization problem, whereas a strong simple control can encode the right structure directly. Preserving every failed iteration made that conclusion possible.

Artifact and ethics statement

No human subjects or personal data were used. The external dataset is EditCLEVR under CC-BY-4.0, pinned by revision; its images are not redistributed in the paper package. The remaining studies use generated data. No model is promoted for deployment. The repository retains negative outcomes and protocol chronology. External submission, endorsement, and independent reproduction are not claimed by the artifact itself.

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